



GTC4Lusso USA - Draining and filling the engine cooling system - Latest Update: 2021-01-26

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B6.07 Draining and filling the engine cooling system

Draining the engine cooling system



CAUTION

The fluid in the engine coolant system may be very hot.

Hot engine coolant may cause severe burns.

ALWAYS WAIT FOR THE SYSTEM TO COOL BEFORE STARTING ANY WORK.



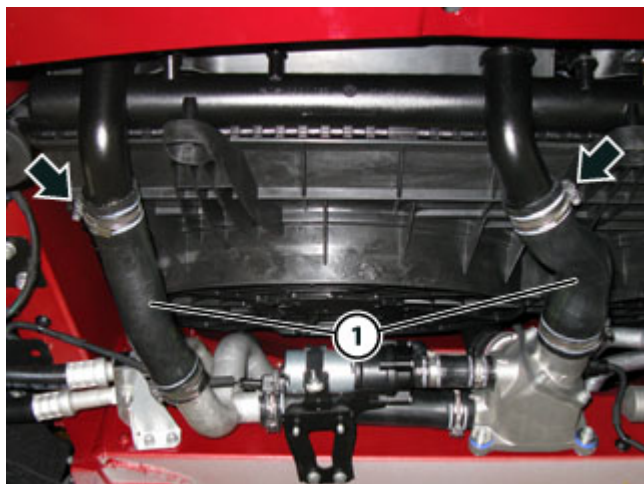
Pour the fluid drained from the system into a container and seal it. DO NOT DISPOSE OF USED FLUID IN THE ENVIRONMENT.



After loosening or removal, the Caillau Visa hose clamps must be replaced with Mikalor worm drive hose clamps, following the instructions given in Technical Information 2054 of December 2012.

➤ Remove the flat undertray sections ( E3.13).

 Remove the front flat undertray section.



- Place a container for collecting engine coolant under the radiator.
- Loosen the indicated screw collars.
- Disconnect the pipes (1).
- Let the engine coolant drain completely.

Filling the engine coolant system

			
Tightening torque		Nm	Class
Fastening engine coolant pipes	Screw collar	5 Nm	B
	Screw collar	4 Nm	B

 After loosening or removal, the Caillau Visa hose clamps must be replaced with Mikalor worm drive hose clamps, following the instructions given in Technical Information 2054 of December 2012.

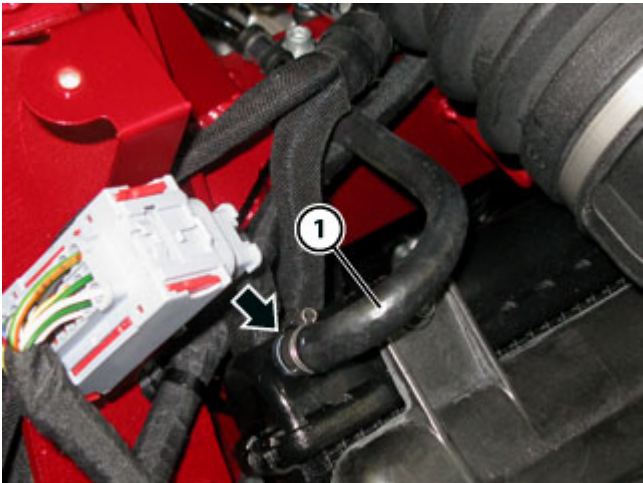
- Remove the engine compartment cosmetic shields ( E3.14).
-  Remove the right hand lateral cosmetic shield.



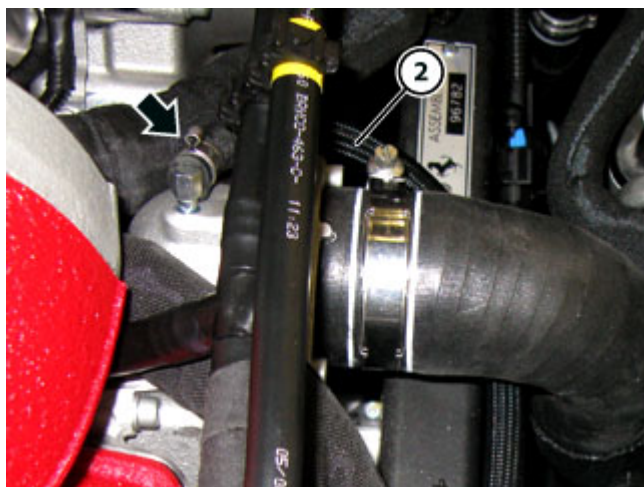
- Connect the pipes (1).
- Tighten the screw collars indicated.

		
Tightening torque	Nm	Class
Screw collar	4 Nm	B

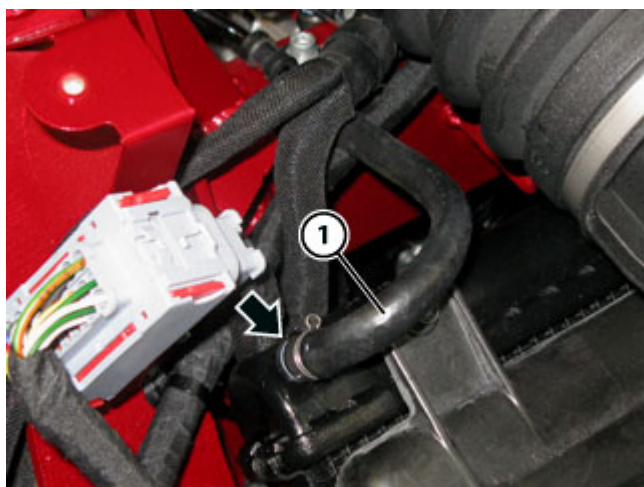
Static filling



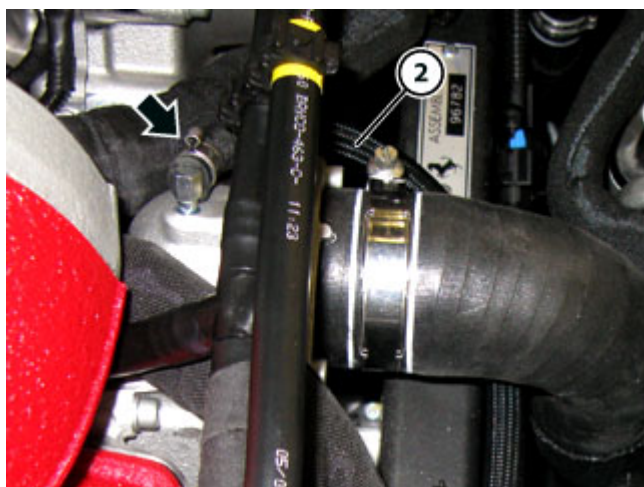
- Loosen the indicated clamp.
- Disconnect the bleed pipe (1) and keep it raised.



- Loosen the indicated clamp.
- Disconnect the bleed pipe (2) and keep it raised.



- Add coolant to the expansion tank until coolant escapes from the radiator bleed union.
- Connect the bleed pipe (1).
- Fit and tighten a new clamp as indicated.



- Add coolant to the expansion tank until coolant escapes from the engine bleed union.
- Connect the bleed pipe (2).
- Fit and tighten a new clamp as indicated.

- Add coolant to the expansion tank until coolant is visible via the filler neck.

Dynamic filling procedure

- Tighten the cap firmly onto the expansion tank.
- Start the engine and run at idle speed for approximately **1 minute**.
- Maintain a steady engine speed of **2000 rpm** for approximately **1 minute**, then allow the engine to idle again.
- Maintain a steady engine speed of **3000 rpm** for approximately **1 minute**, then allow the engine to idle again.
- Check via the filler neck that engine coolant is visible in the reservoir.
- If the coolant is not visible via the filler neck, top up with the specified coolant until the level is visible.
- Maintain a steady engine speed of **2000 rpm** until the cooling fans are activated (engine temperature approx. **95°C**), then allow the engine to idle again.
- Check via the filler neck that engine coolant is visible in the reservoir.
- If the coolant is not visible via the filler neck, top up with the specified coolant until the level is visible.
- Set the AC system to the maximum temperature for one cabin zone only to open the relative heater circuit.
- Maintain a steady engine speed of **3000 rpm** for approximately **1 minute**, then allow the engine to idle again for approximately **30 seconds**.
- Return the cabin zone set to the maximum temperature to the previous setting and set the other zone to the maximum temperature to open the relative heater circuit.
- Maintain a steady engine speed of **3000 rpm** for approximately **1 minute**, then allow the engine to idle again for approximately **30 seconds**.
- Close the heater circuit.
- Check via the filler neck that engine coolant is visible in the reservoir.
- If the coolant is not visible via the filler neck, top up with the specified coolant until the level is visible.
- Repeat the seven steps described previously in the correct sequence.
- Set the AC system to the maximum temperature for both cabin zones to open the heater circuit.
- Maintain a steady engine speed of **3000 rpm** for approximately **1 minute**, then allow the engine to idle again for approximately **30 seconds**.
- Close the heater circuit.
- Maintain a steady engine speed of **3000 rpm** for approximately **1 minute**, then allow the engine to idle again for approximately **30 seconds**.
- Switch off the engine.
- Unscrew and open the expansion tank cap.
- Wait (approx. 1 hour) for the engine temperature to cool to **40-45°C**.
- Check via the filler neck that engine coolant is visible in the reservoir.
- If the coolant is not visible via the filler neck, top up with the specified coolant until the level is visible.
- Tighten the cap firmly onto the expansion tank.

➤ Refit the engine compartment cosmetic shields ( E3.14).

 Refit the right hand lateral cosmetic shield.

➤ Refit the flat undertray sections ( E3.13).

 Refit the front flat undertray section.